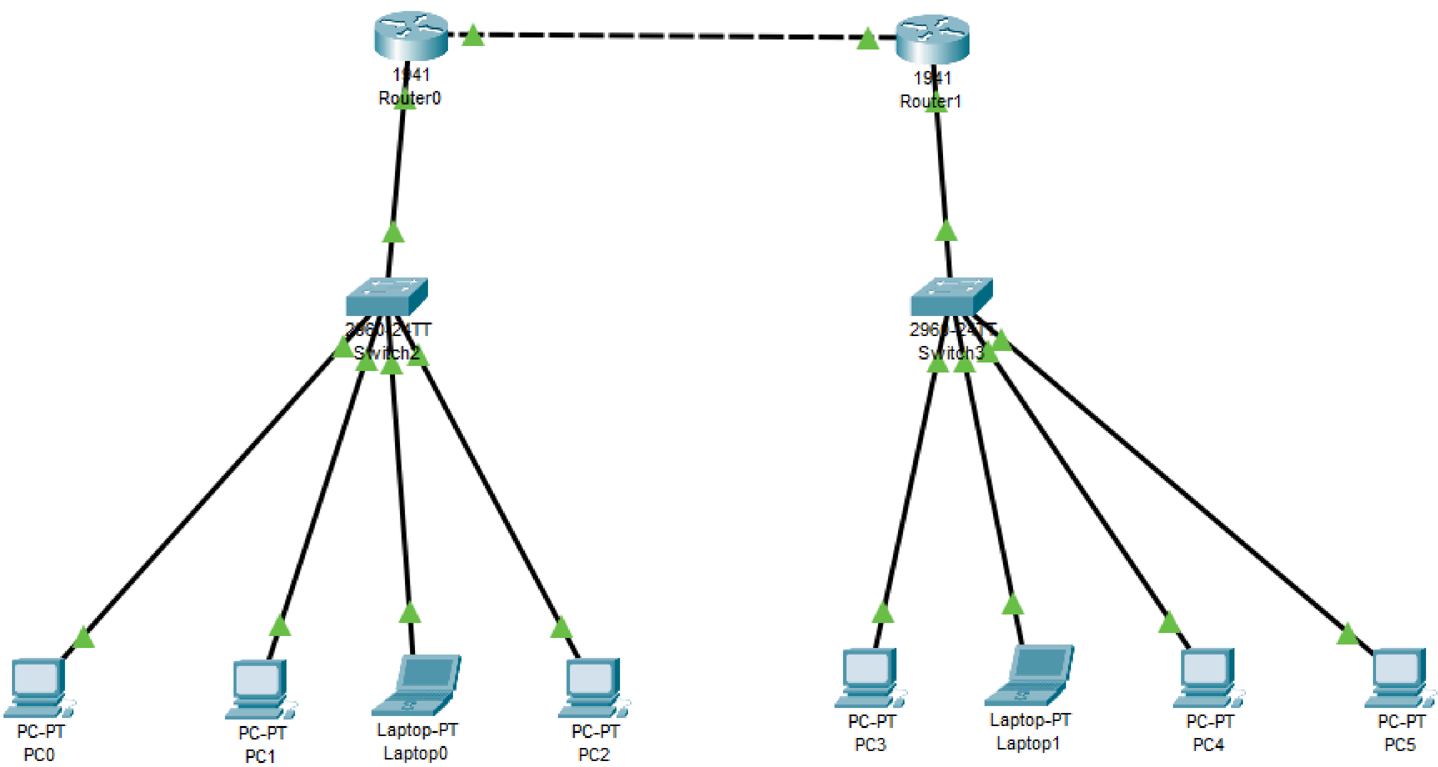
Computer Networks

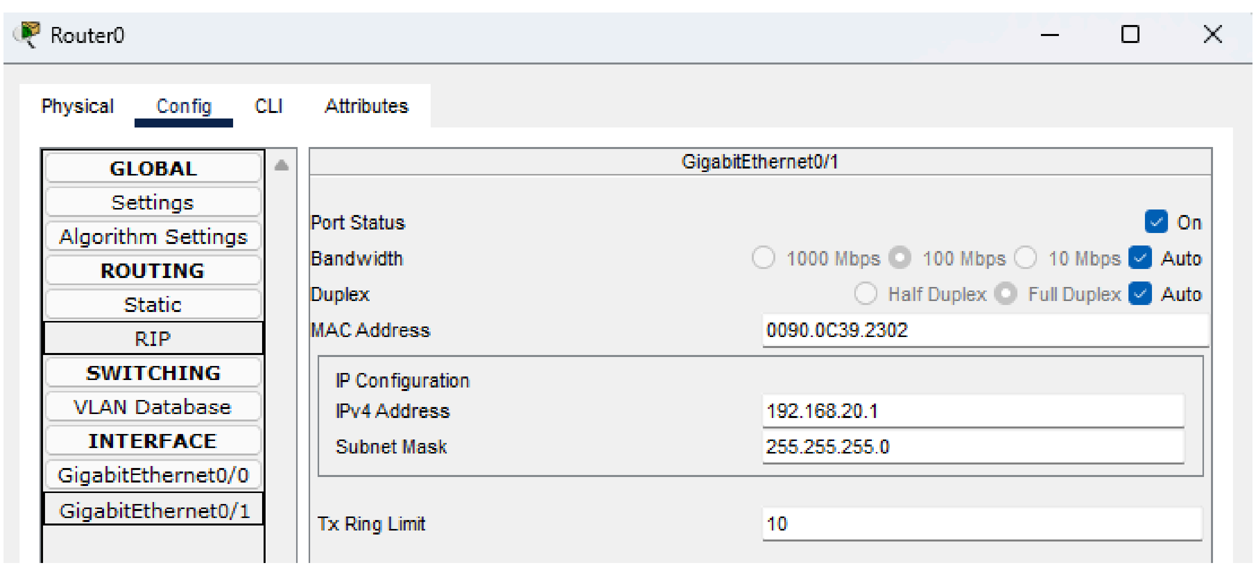
LAB#10

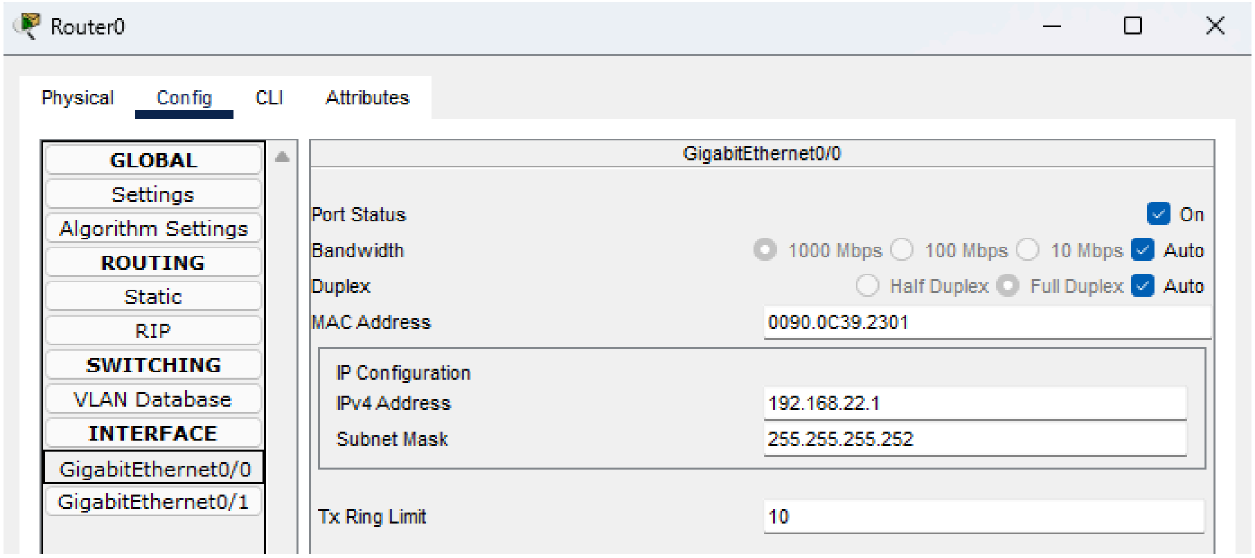
23K-2001

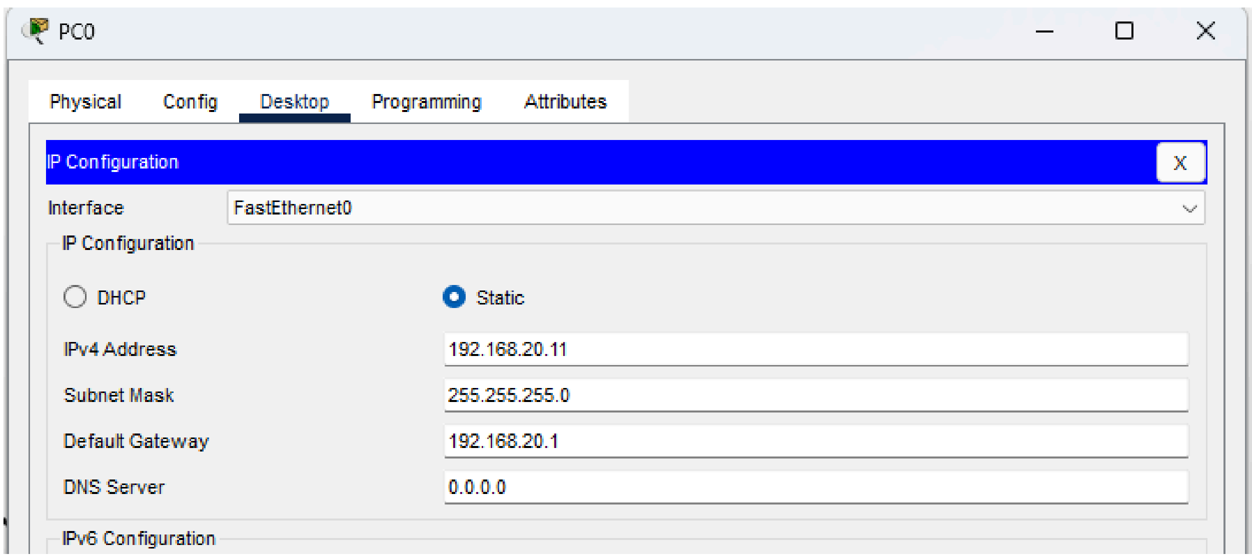
BCS-5J

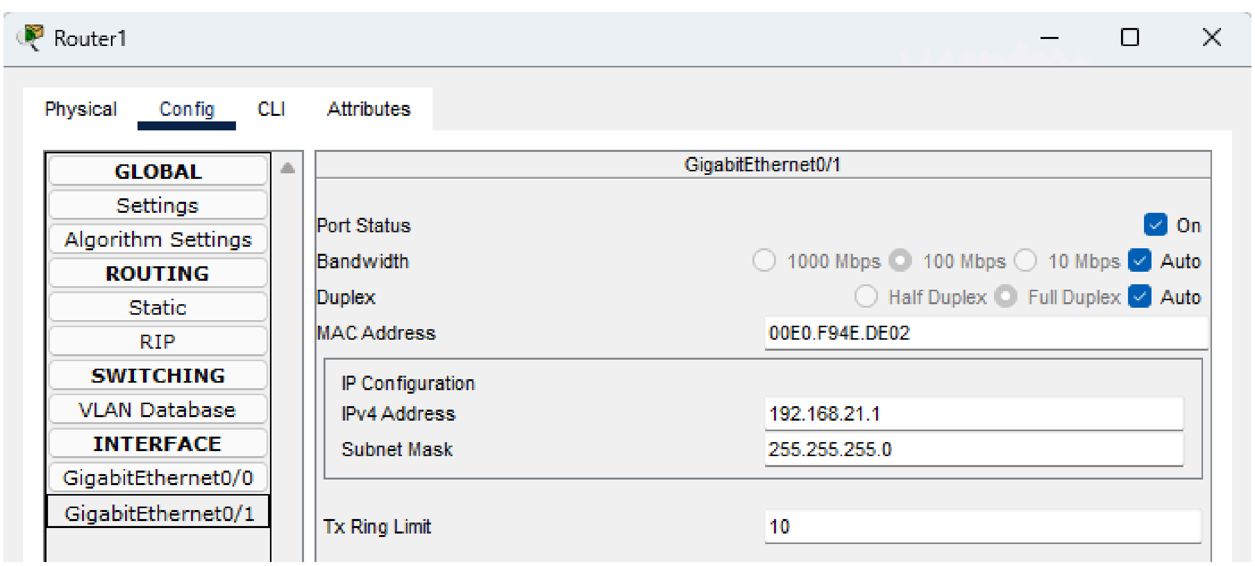
Q1:

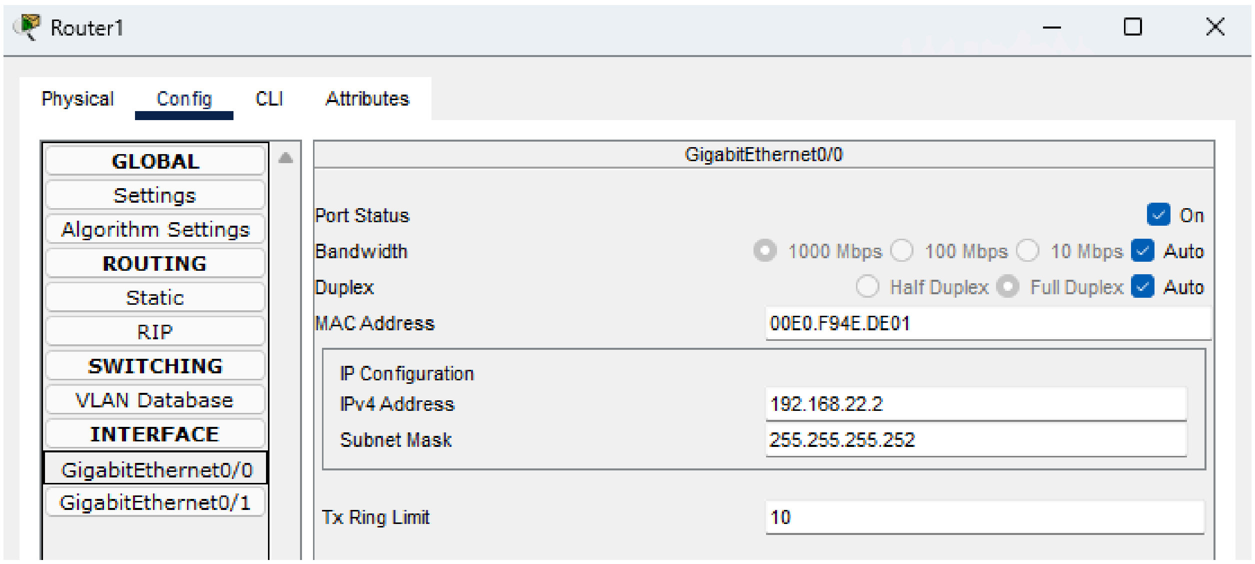

1941
Router0
1941
Router1
2960-24TT
Switch2
2960-24TT
Switch3
PC-PT
PC0
PC-PT
PC1
Laptop-PT
Laptop0
PC-PT
PC2
PC-PT
PC3
Laptop-PT
Laptop1
PC-PT
PC4
PC-PT
PC5


Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/1
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0090.0C39.2302
IP Configuration
IPv4 Address
192.168.20.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

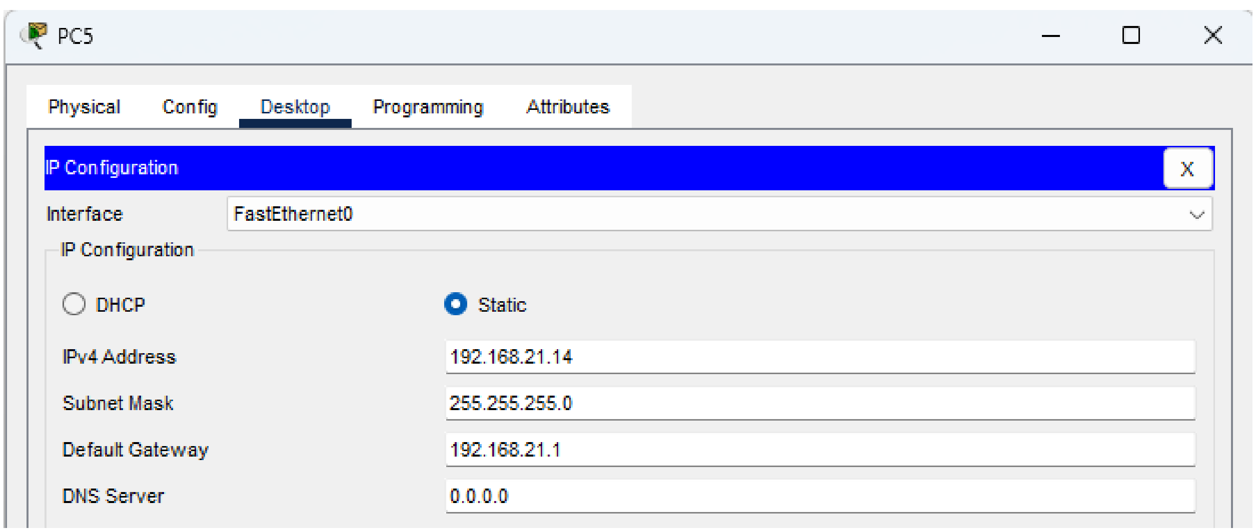

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0090.0C39.2301
IP Configuration
IPv4 Address
192.168.22.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
0.0.0.0
IPv6 Configuration


Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/1
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.F94E.DE02
IP Configuration
IPv4 Address
192.168.21.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10



Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00E0.F94E.DE01
IP Configuration
IPv4 Address
192.168.22.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10



PC5
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.21.14
Subnet Mask
255.255.255.0
Default Gateway
192.168.21.1
DNS Server
0.0.0.0

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wwl/export/crypto/tool/stqrg.html

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up


Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ex
Router(config)#router ospf 1
Router(config-router)#network 192.168.21.0 0.0.0.255 area 0
Router(config-router)#network 192.168.22.0 0.0.0.3 area 0
Router(config-router)#
00:22:37: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.22.1 on GigabitEthernet0/0 from LOADING
to FULL, Loading Done
```

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Top

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ex
Router(config)#router ospf 1
Router(config-router)#network 192.168.20.0 0.0.0.255 area 0
Router(config-router)#network 192.168.
                                      ^
% Invalid input detected at '^' marker.

Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/1
Router(config-if)#
%SYS-5-CONFIG_I: Configured from console by console

Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ex
Router(config)#router ospf 1
Router(config-router)#network 192.168.22.0 0.0.0.3 area 0
Router(config-router)#
00:22:37: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.22.2 on GigabitEthernet0/0 from LOADING
to FULL, Loading Done
```

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Top

Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Router>en
Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.22.2      1   FULL/DR         00:00:37    192.168.22.2    GigabitEthernet0/0
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/1
L       192.168.20.1/32 is directly connected, GigabitEthernet0/1
O    192.168.21.0/24 [110/2] via 192.168.22.2, 00:16:57, GigabitEthernet0/0
     192.168.22.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.22.0/30 is directly connected, GigabitEthernet0/0
L       192.168.22.1/32 is directly connected, GigabitEthernet0/0

Router#
```

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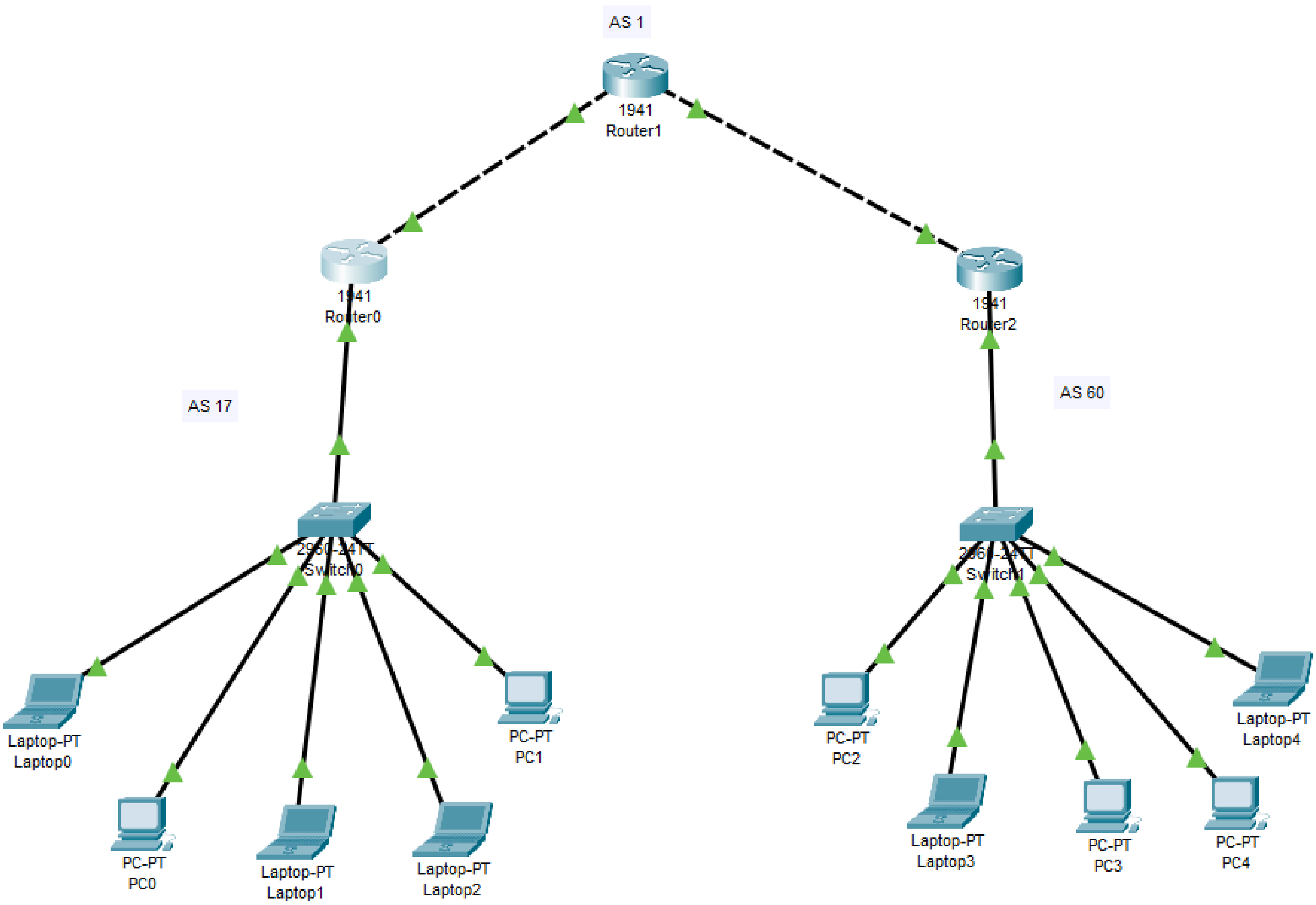
Router1

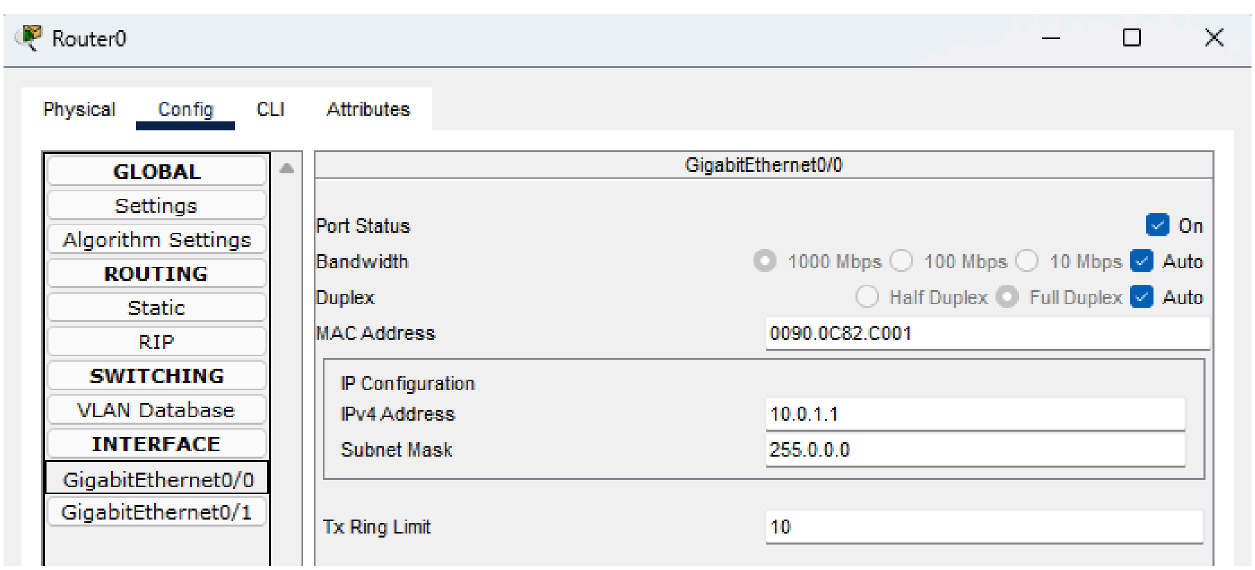
Physical Config CLI Attributes

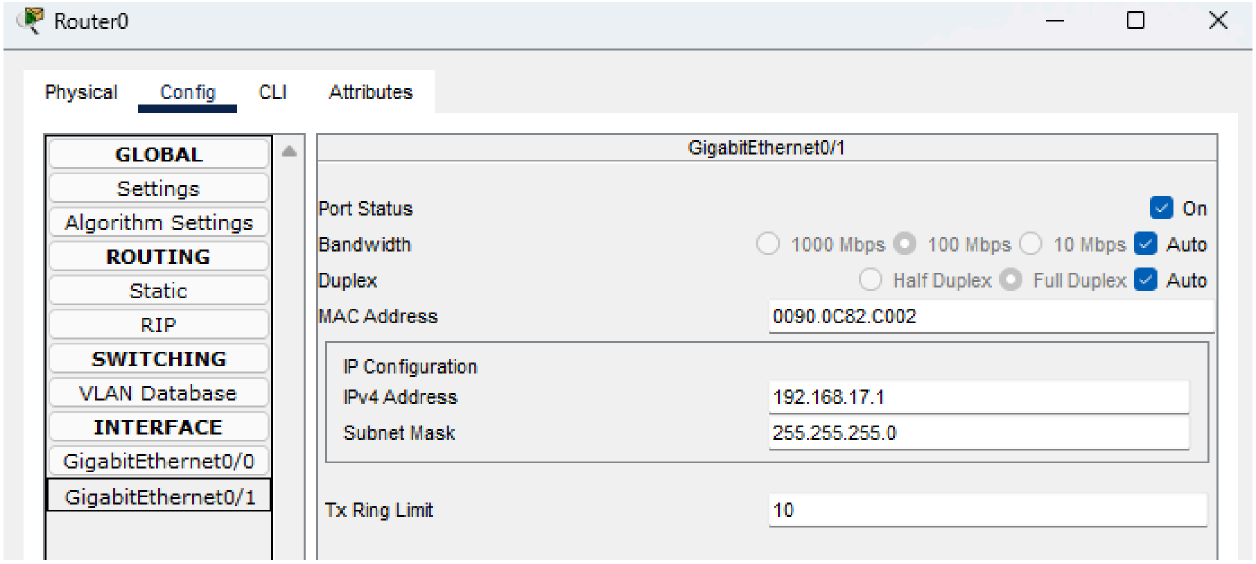
IOS Command Line Interface

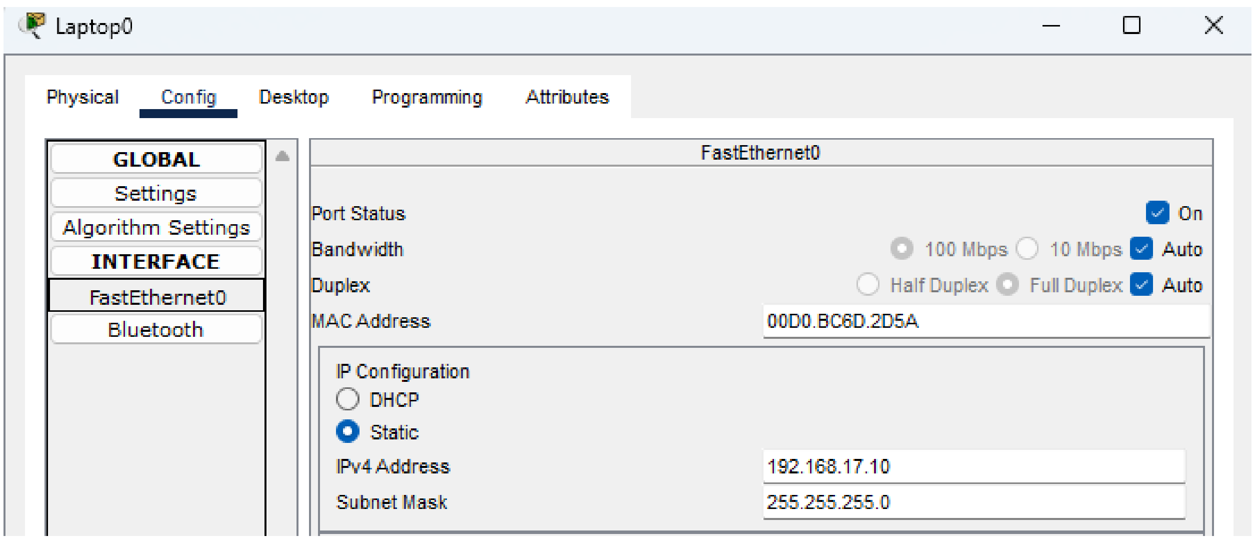
```
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip ospf neighbor


Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.22.1      1   FULL/DR         00:00:38    192.168.22.1    GigabitEthernet0/0
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    192.168.20.0/24 [110/2] via 192.168.22.1, 00:02:56, GigabitEthernet0/0
     192.168.21.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.21.0/24 is directly connected, GigabitEthernet0/1
L       192.168.21.1/32 is directly connected, GigabitEthernet0/1
     192.168.22.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.22.0/30 is directly connected, GigabitEthernet0/0
L       192.168.22.2/32 is directly connected, GigabitEthernet0/0

Router#
```

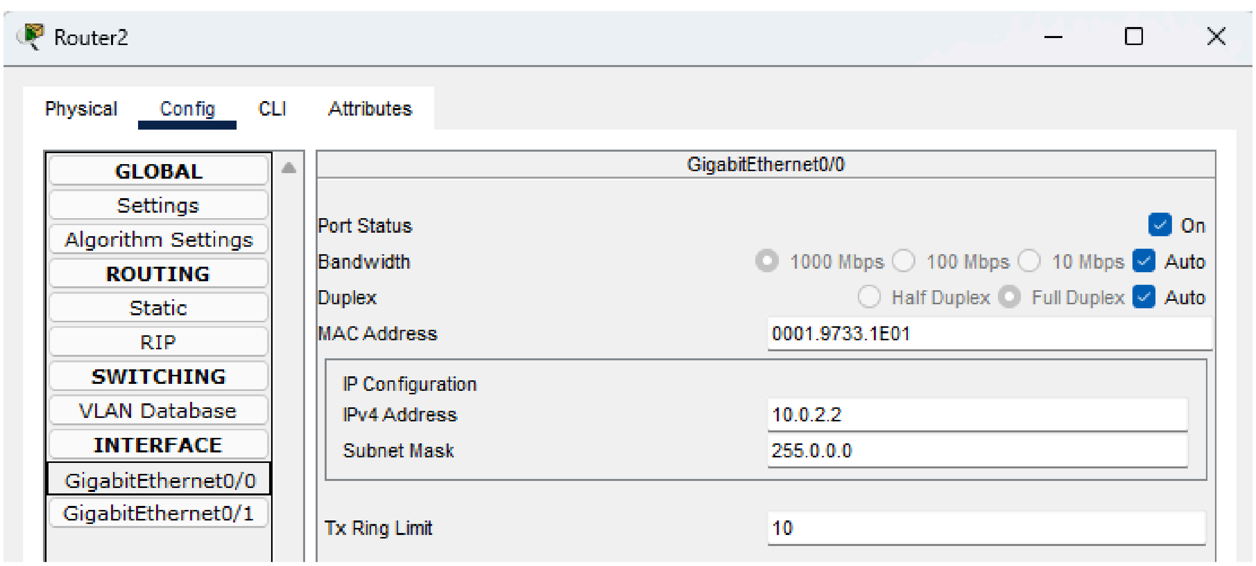
Q2:

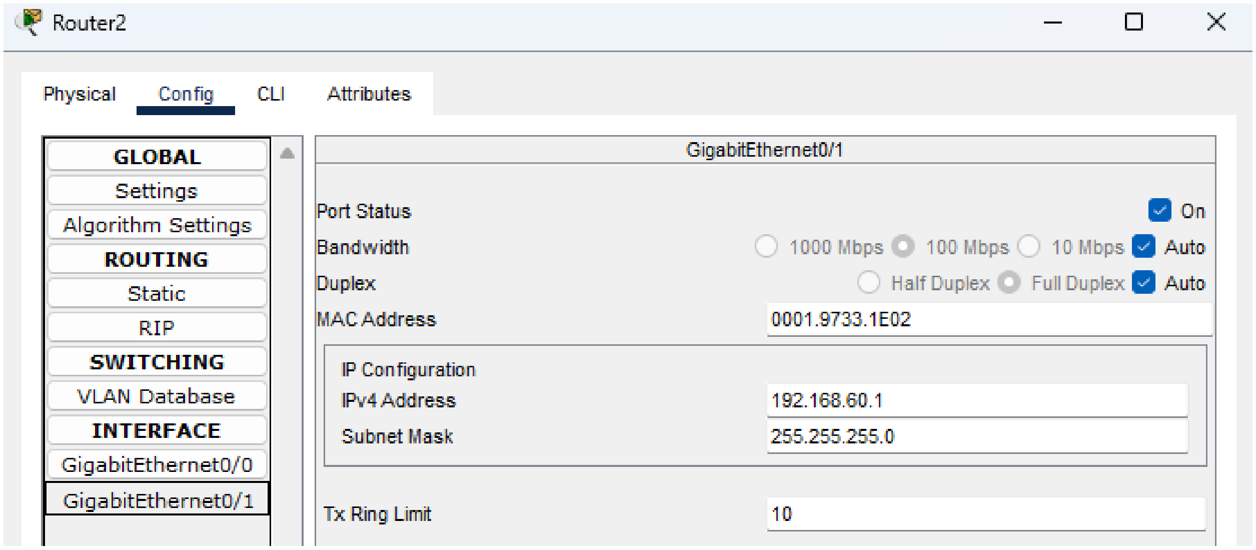

AS 1
1941
Router1
1941
Router0
1941
Router2
AS 17
AS 60
Laptop-PT
Laptop0
PC-PT
PC0
Laptop-PT
Laptop1
Laptop-PT
Laptop2
PC-PT
PC1
PC-PT
PC2
Laptop-PT
Laptop3
PC-PT
PC3
PC-PT
PC4
Laptop-PT
Laptop4

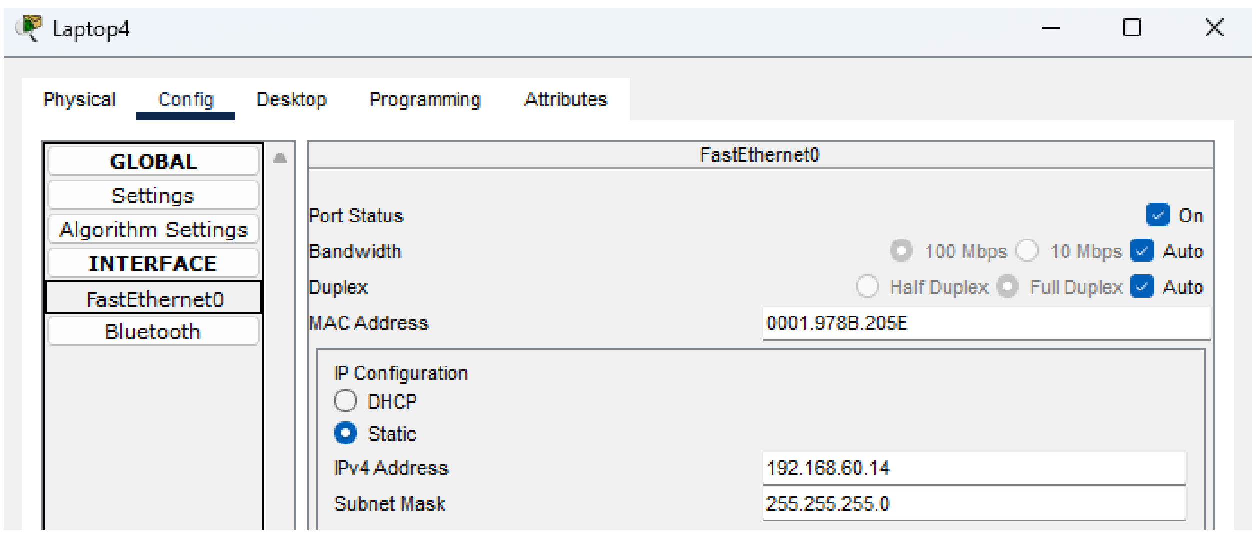

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0090.0C82.C001
IP Configuration
IPv4 Address
10.0.1.1
Subnet Mask
255.0.0.0
Tx Ring Limit
10

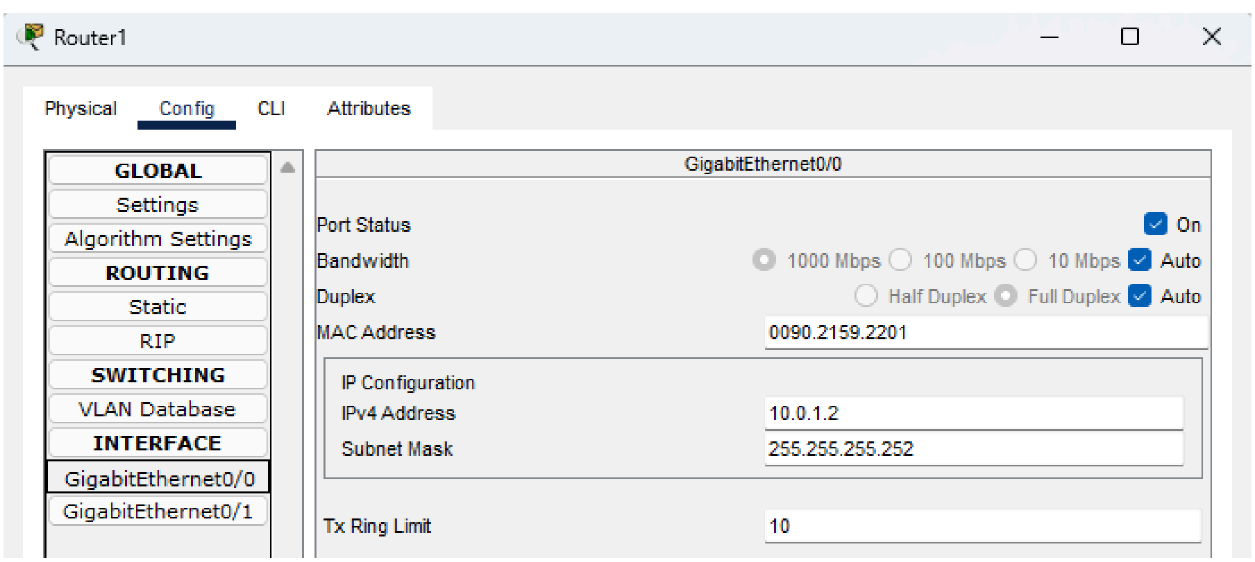

Router0
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/1
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0090.0C82.C002
IP Configuration
IPv4 Address
192.168.17.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10

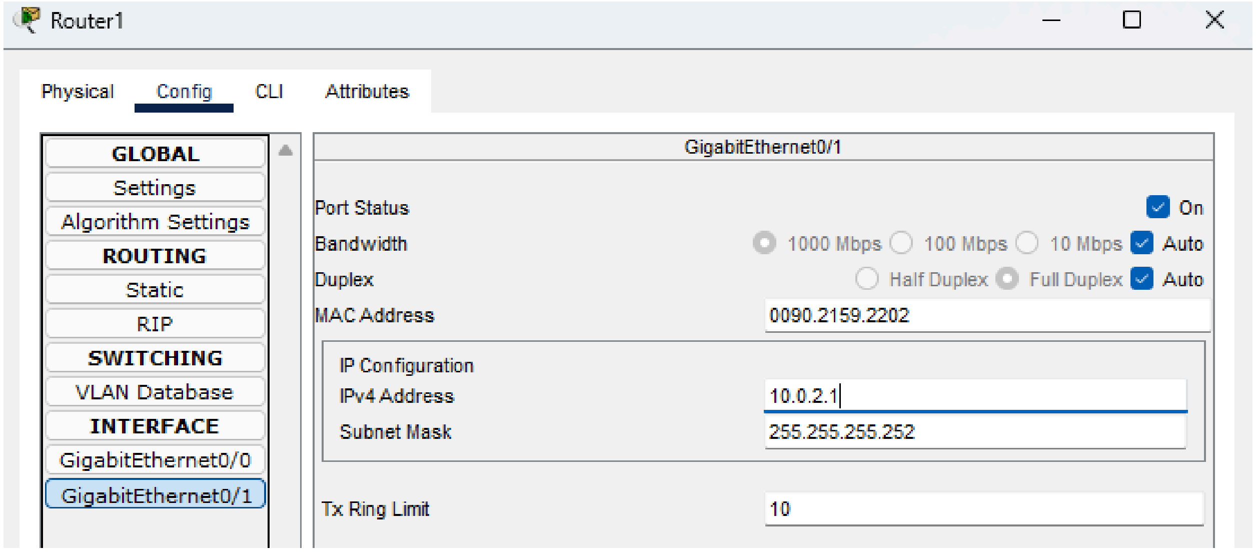

Laptop0
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
00D0.BC6D.2D5A
IP Configuration
DHCP
Static
IPv4 Address
192.168.17.10
Subnet Mask
255.255.255.0

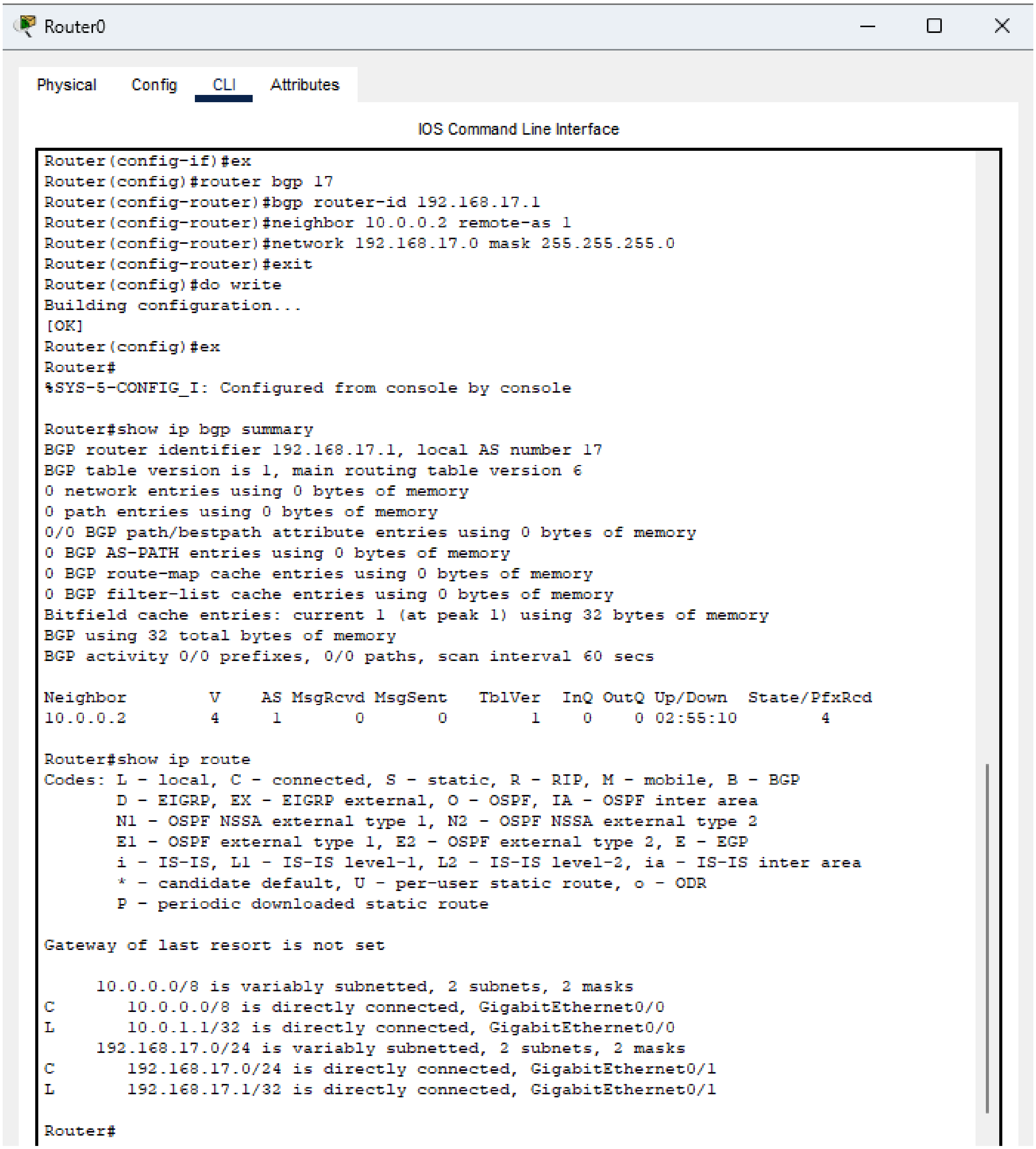

Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.9733.1E01
IP Configuration
IPv4 Address
10.0.2.2
Subnet Mask
255.0.0.0
Tx Ring Limit
10


Router2
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/1
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.9733.1E02
IP Configuration
IPv4 Address
192.168.60.1
Subnet Mask
255.255.255.0
Tx Ring Limit
10


Laptop4
Physical
Config
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Bluetooth
FastEthernet0
Port Status
On
Bandwidth
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0001.978B.205E
IP Configuration
DHCP
Static
IPv4 Address
192.168.60.14
Subnet Mask
255.255.255.0


Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/0
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0090.2159.2201
IP Configuration
IPv4 Address
10.0.1.2
Subnet Mask
255.255.255.252
Tx Ring Limit
10


Router1
Physical
Config
CLI
Attributes
GLOBAL
Settings
Algorithm Settings
ROUTING
Static
RIP
SWITCHING
VLAN Database
INTERFACE
GigabitEthernet0/0
GigabitEthernet0/1
GigabitEthernet0/1
Port Status
On
Bandwidth
1000 Mbps
100 Mbps
10 Mbps
Auto
Duplex
Half Duplex
Full Duplex
Auto
MAC Address
0090.2159.2202
IP Configuration
IPv4 Address
10.0.2.1
Subnet Mask
255.255.255.252
Tx Ring Limit
10


Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config-if)#ex
Router(config)#router bgp 17
Router(config-router)#bgp router-id 192.168.17.1
Router(config-router)#neighbor 10.0.0.2 remote-as 1
Router(config-router)#network 192.168.17.0 mask 255.255.255.0
Router(config-router)#exit
Router(config)#do write
Building configuration...
[OK]
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#show ip bgp summary
BGP router identifier 192.168.17.1, local AS number 17
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs
Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State/PfxRcd
10.0.0.2 4 1 0 0 1 0 0 02:55:10 4
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C 10.0.0.0/8 is directly connected, GigabitEthernet0/0
L 10.0.1.1/32 is directly connected, GigabitEthernet0/0
192.168.17.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.17.0/24 is directly connected, GigabitEthernet0/1
L 192.168.17.1/32 is directly connected, GigabitEthernet0/1
Router#

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#ex
Router(config)#router bgp 1
Router(config-router)#bgp router-id 10.0.0.2
Router(config-router)#neighbor 10.0.0.1 remote-as 17
Router(config-router)#neighbor 10.0.0.6 remote-as 60
Router(config-router)#exit
Router(config)#do write
Building configuration...
[OK]
Router(config)#ex
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip bgp summary
BGP router identifier 10.0.0.2, local AS number 1
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.1        4    17       0       0        1    0    0 02:54:53        4
10.0.0.6        4    60       0       0        1    0    0 02:54:53        4

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.1.0/30 is directly connected, GigabitEthernet0/0
L       10.0.1.2/32 is directly connected, GigabitEthernet0/0
C       10.0.2.0/30 is directly connected, GigabitEthernet0/1
L       10.0.2.1/32 is directly connected, GigabitEthernet0/1

Router#
```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#ex
Router(config)#router bgp 60
Router(config-router)#bgp router-id 192.168.60.1
Router(config-router)#neighbor 10.0.0.5 remote-as 1
Router(config-router)#network 192.168.60.0 mask 255.255.255.0
Router(config-router)#exit
Router(config)#do write
Building configuration...
[OK]
```

Router2

Physical Config CLI Attributes

IOS Command Line Interface

```
Router#show ip bgp summary
BGP router identifier 192.168.60.1, local AS number 60
BGP table version is 1, main routing table version 6
0 network entries using 0 bytes of memory
0 path entries using 0 bytes of memory
0/0 BGP path/bestpath attribute entries using 0 bytes of memory
0 BGP AS-PATH entries using 0 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 32 total bytes of memory
BGP activity 0/0 prefixes, 0/0 paths, scan interval 60 secs

Neighbor        V    AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.0.0.5        4     1       0       0        1    0    0 02:54:30        4

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/0
L       10.0.2.2/32 is directly connected, GigabitEthernet0/0
     192.168.60.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.60.0/24 is directly connected, GigabitEthernet0/1
L       192.168.60.1/32 is directly connected, GigabitEthernet0/1

Router#
```